Signals and Systems Lecture-9: Fourier Series and LTI Systems

Chadi Abou-Rjeily

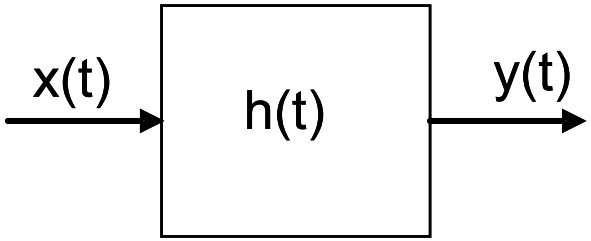
Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

January 31, 2023

LTI systems: Impulse Response

- We have previously seen that Linear and Time-Invariant (LTI) systems are characterized by the following input-output relation:

$$y(t) = h(t) * x(t)$$

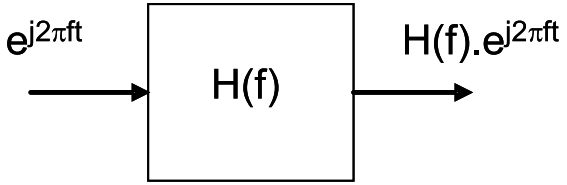


where:

- $x(t)$ is the input.
- $y(t)$ is the output.
- $h(t)$ is the impulse response of the system.

LTI systems: Transfer function (1)

- Instead of characterizing a LTI system by its impulse response, this system can be characterized in an equivalent way by its **frequency response** or **transfer function** $H(f)$ such that:



- Following from the linearity of the system, if the input is written as:

$$x(t) = \sum_{k=-\infty}^{+\infty} X[k] e^{j2\pi(kf_0)t}$$

then the output of the LTI system will be written as:

$$y(t) = \sum_{k=-\infty}^{+\infty} X[k] \left[H(kf_0) e^{j2\pi(kf_0)t} \right] = \sum_{k=-\infty}^{+\infty} \underbrace{[X[k]H(kf_0)]}_{=Y[k]} e^{j2\pi(kf_0)t}$$

LTI systems: Transfer function (2)

- Consequently:

$$X[k] \quad \rightarrow \quad Y[k] = \underbrace{H(kf_0)}_{\text{“gain”}} X[k]$$

  - The FS coefficients of the output are given by: $Y[k] = H(kf_0)X[k]$.
  - Note that the integer k corresponds to the frequency kf_0 (where f_0 is the fundamental frequency).
- Note that $H(kf_0)$ includes both a magnitude and a phase:

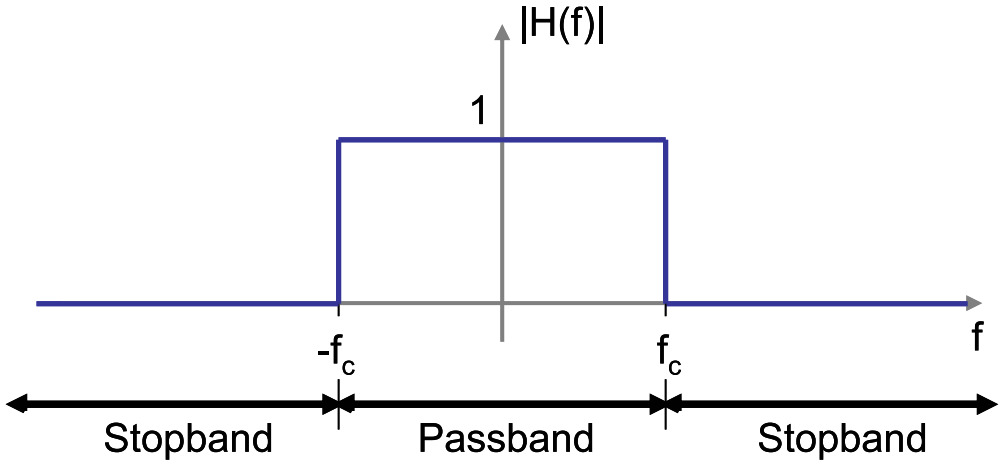
$$H(kf_0) = |H(kf_0)|e^{j\arg[H(kf_0)]}$$

- In particular:

$$\begin{aligned} |X[k]| \quad &\rightarrow \quad |Y[k]| = |H(kf_0)||X[k]| \\ \arg[X[k]] \quad &\rightarrow \quad \arg[Y[k]] = \arg[H(kf_0)] + \arg[X[k]] \end{aligned}$$

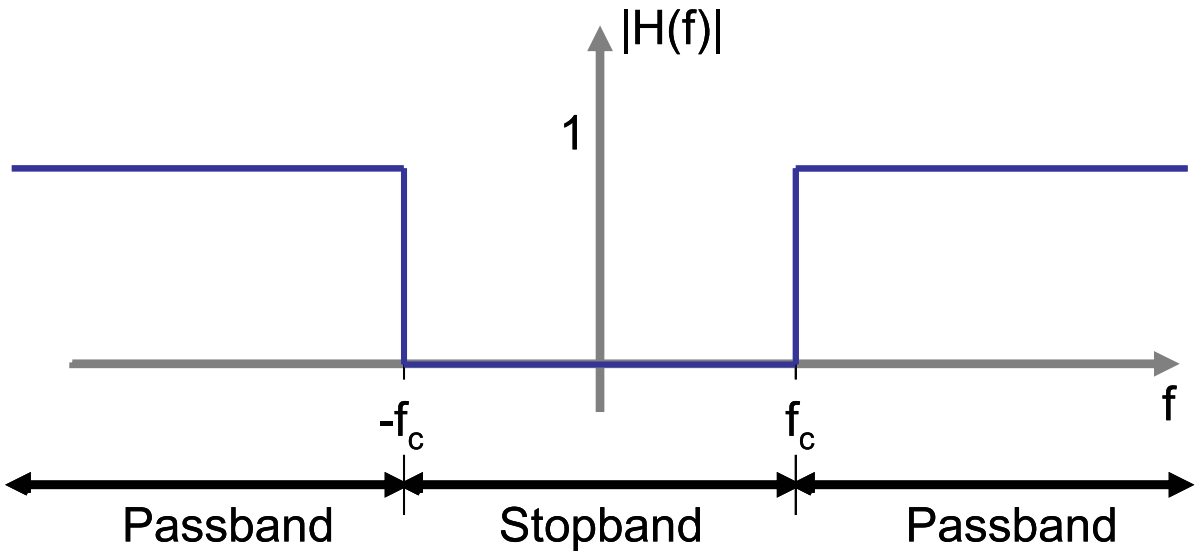
Filters (1)

- By a convenient choice of $H(f)$ as a function of f, we can shape the frequency composition of the input:
  - Preferential Amplification.
  - Selective filtering of some frequencies.
- Ideal Lowpass filter: permits to eliminate the frequencies that are **not** included in $[-f_c \;\; f_c]$ where f_c is called the cutoff frequency.



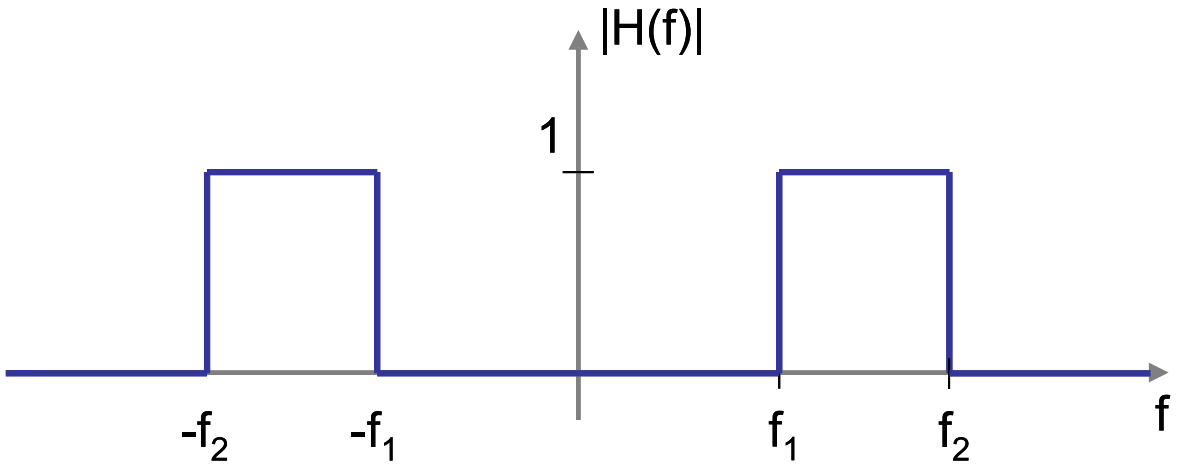
Filters (2)

- Ideal Highpass filter: permits to eliminate the frequencies that are included in $[-f_c \;\; f_c]$ where f_c is called the cutoff frequency.



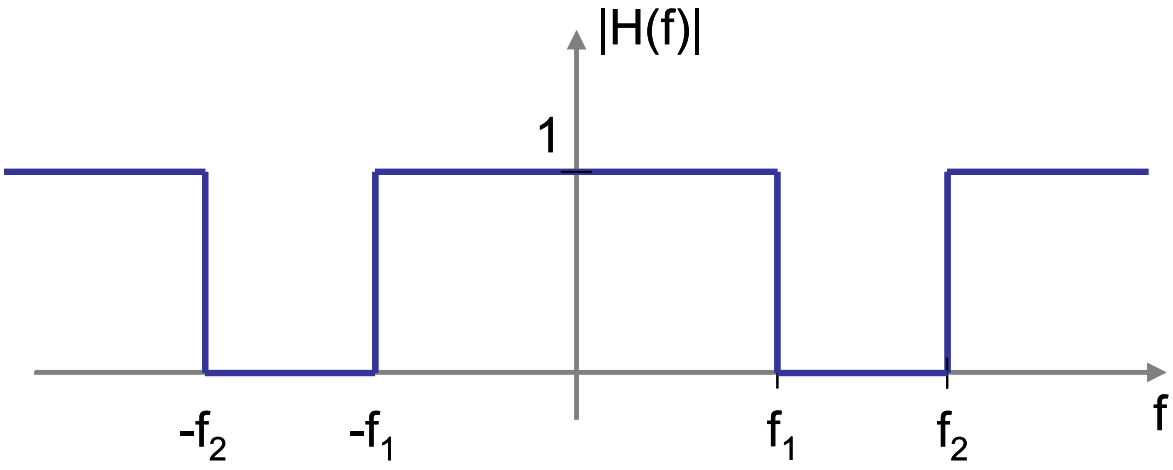
Filters (3)

- Ideal Bandpass filter: permits to eliminate the frequencies that are **not** included in the intervals $[f_1 \quad f_2]$ and $[-f_2 \quad -f_1]$ where f_1 is the lower cutoff frequency and f_2 is the upper cutoff frequency.



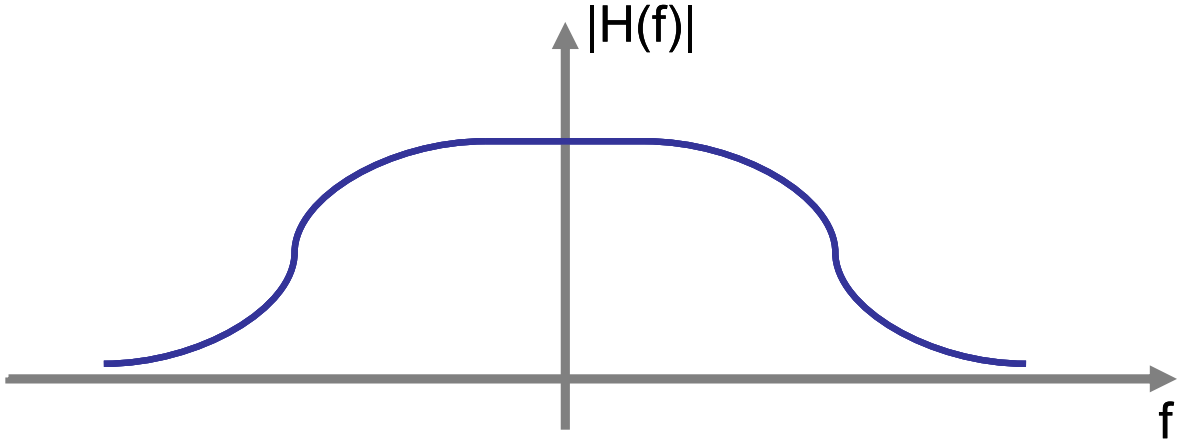
Filters (4)

- Ideal Bandstop filter: permits to eliminate the frequencies that are included in the intervals $[f_1 \quad f_2]$ and $[-f_2 \quad -f_1]$ where f_1 is the lower cutoff frequency and f_2 is the upper cutoff frequency.



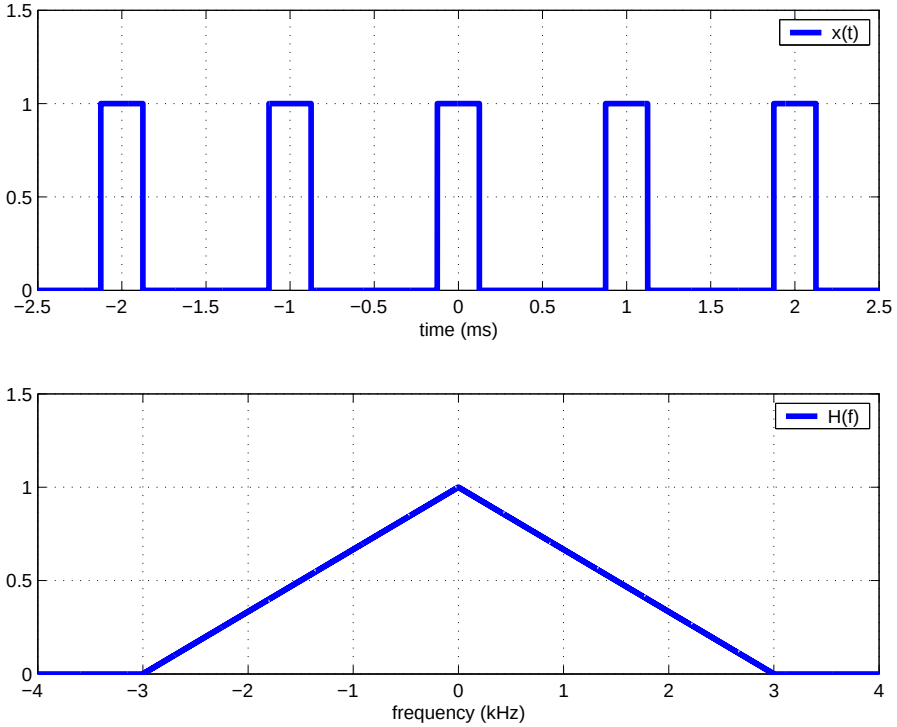
Filters (5)

- In general, the filters are not ideal. They are not capable of completely filtering out the signals outside the frequency range of interest.
- Example: A non-ideal Lowpass filter:



Example-1 (1)

The signal $x(t)$ shown in the figure below is applied to the input of a non-ideal low-pass filter whose transfer function is given by $H(f) = \mathrm{tri}\left(\frac{f}{3000}\right)$. Write down the expression of the signal $y(t)$ at the output of the filter and calculate the percentage of the input power that appears at the output of this filter.



Example-1 (2)

- $x(t)$ corresponds to the repetition of a rectangular pulse of duration $w = 0.25$ ms with a period $T_0 = 1$ ms.
- Consequently, the FS coefficients of $x(t)$ are given by:

$$X[k] = \frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) = \frac{1}{4}\text{sinc}\left(\frac{k}{4}\right)$$

- The fundamental frequency of $x(t)$ is $f_0 = \frac{1}{T_0} = 1$ kHz $= 1000$ Hz. Since $H(f) = \text{tri}\left(\frac{f}{3000}\right)$ is nonzero only in the interval $[-3000 \;\; 3000]$ Hz, then the frequency components that will pass through the filter must satisfy the following relation:

$$\begin{aligned} & -3000 < kf_0 < 3000 \\ \Rightarrow \quad & -3000 < 1000k < 3000 \quad \Rightarrow \quad k = 0, \pm 1, \pm 2 \end{aligned}$$

Example-1 (3)

- On the other hand, $\mathrm{tri}(f) = 1 - |f|$ implying that:

$$H(f) = \mathrm{tri}\left(\frac{f}{3000}\right) = 1 - \left|\frac{f}{3000}\right| \quad (f \text{ in Hz})$$

- The FS coefficients of the output signal are given by:

$$Y[0] = H(0 \times 1000)X[0] = 1\left[\frac{1}{4}\mathrm{sinc}\left(\frac{0}{4}\right)\right] = 0.25$$

$$Y[1] = H(1 \times 1000)X[1] = \left[1 - \frac{1}{3}\right]\left[\frac{1}{4}\mathrm{sinc}\left(\frac{1}{4}\right)\right] \simeq 0.15$$

$$Y[2] = H(2 \times 1000)X[2] = \left[1 - \frac{2}{3}\right]\left[\frac{1}{4}\mathrm{sinc}\left(\frac{2}{4}\right)\right] \simeq 0.053$$

- Note that $X[-k] = X[k]$ (the sinc function is even) and $H(-f) = H(f)$ (the tri function is even). Consequently, $Y[-k] = Y[k]$.

Example-1 (4)

- As a conclusion, for the output signal $y(t)$:

$$
\begin{aligned}
&Y[0] = 0.25 \\
&Y[-1] = Y[1] = 0.15 \\
&Y[-2] = Y[2] = 0.053 \\
&Y[k] = 0 \;\; ; \;\; k \neq 0, \pm 1, \pm 2
\end{aligned}
$$

- Consequently:

$$
\begin{aligned}
y(t) &= 0.25 + 0.15 \left[e^{j2\pi(1000)t} + e^{j2\pi(-1000)t} \right] \\
&\quad + 0.053 \left[e^{j2\pi(2000)t} + e^{j2\pi(-2000)t} \right] \\
&= 0.25 + 0.3 \cos(2000\pi t) + 0.106 \cos(4000\pi t)
\end{aligned}
$$

Note that $y(t)$ is real and even and its FS coefficients are real (as expected).

Example-1 (5)

- Applying Parseval's theorem, the average power of the output signal is given by ($y(t)$ is real):

$$P_{out} = |Y[0]|^2 + 2\left[|Y[1]|^2 + |Y[2]|^2\right]$$
$$= (0.25)^2 + 2\left[(0.15)^2 + (0.053)^2\right] \simeq 0.1131 \quad (\mathrm{W})$$

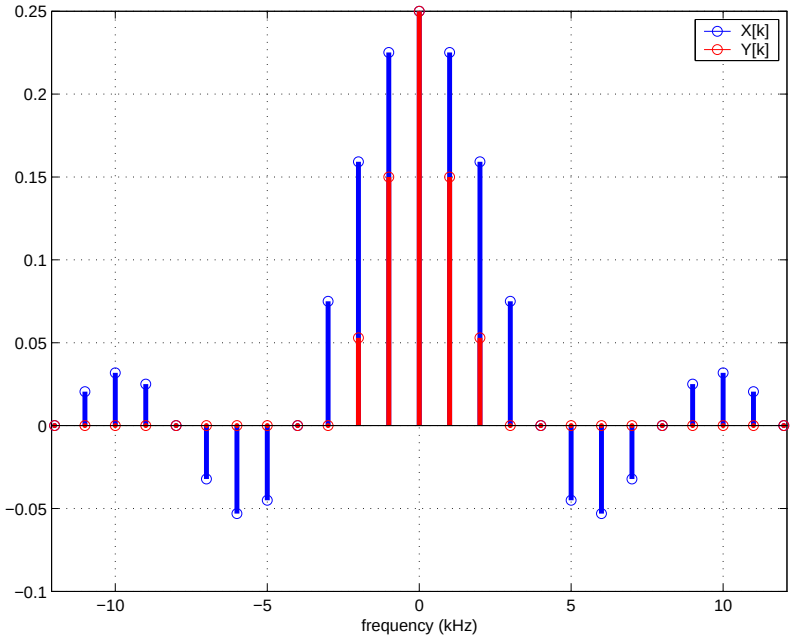
- Since $x(t)$ has an infinite number of FS coefficients, it is more convenient to calculate its power in the time domain:

$$P_{\text{in}} = \frac{1}{T_0}\int_{T_0} x^2(t)\mathrm{d}t = \frac{1}{1}\int_{-0.5}^{0.5} x^2(t)\mathrm{d}t \quad (T_0 \text{ in ms})$$
$$= \int_{-0.5}^{-0.125} 0^2\mathrm{d}t + \int_{-0.125}^{0.125} 1^2\mathrm{d}t + \int_{0.125}^{0.5} 0^2\mathrm{d}t = 0.25 \ \ \mathrm{W}$$

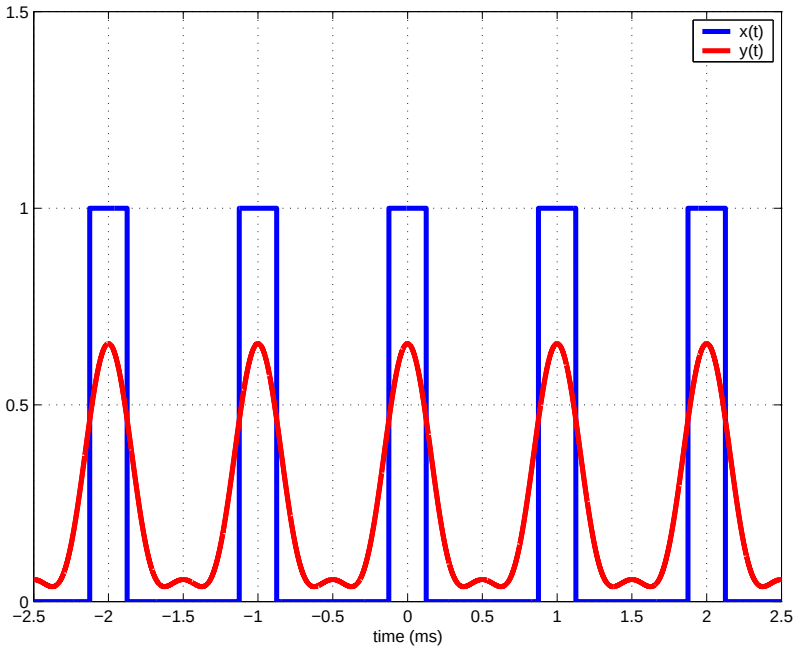
- Consequently, $\frac{P_{out}}{P_{in}} \times 100\% = 45.24\%$ of the input power appears at the output of the low-pass filter.

Example-1 (6)

- The spectra of the input and output signals are shown in the figure below:



- The input and output signals $x(t)$ and $y(t)$ are shown in the figure below:

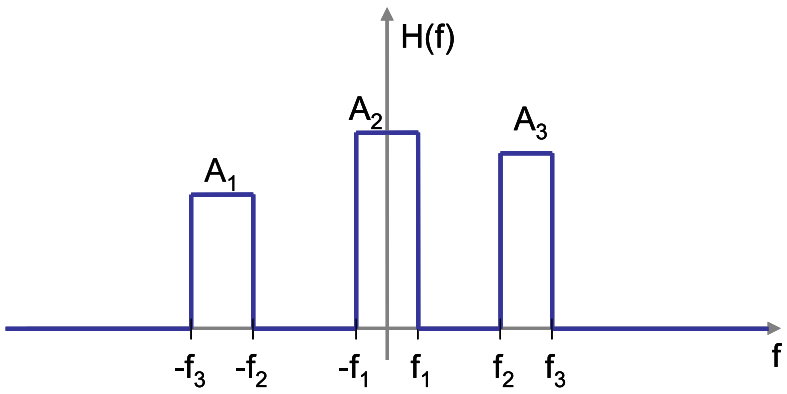


Example-2 (1)

- Consider the periodic signal with period $T_0 = 4$ whose exponential FS coefficients are given by:

$$X[k] = \begin{cases} 2\frac{\sin\left(k\frac{\pi}{2}\right)}{j(k\pi)^2}e^{-jk\frac{\pi}{2}}, & k \neq 0; \\ \frac{1}{2}, & k = 0. \end{cases}$$

and consider the LTI system whose frequency response $H(f)$ is shown below:



- Determine the values of A_1, A_2, A_3, f_1, f_2 and f_3 so that the response of this system to $x(t)$ is equal to:

$$y(t) = 1 - \cos\left(\frac{3\pi}{2}t\right)$$

Example-2 (2)

- $T_0 = 4 \Rightarrow f_0 = \frac{1}{4}$ and consequently:

$$3\frac{\pi}{2} = 2\pi\left(\frac{3}{4}\right) = 2\pi\,(3f_0)$$

- Consequently, in terms of the fundamental frequency f_0, $y(t)$ can be written as:

$$y(t) = 1 - \cos(2\pi(3f_0)t) = 1 - \frac{1}{2}e^{j2\pi(-3f_0)t} - \frac{1}{2}e^{j2\pi(3f_0)t}$$

- Implying that the FS coefficients of the output are given by:

$$\begin{aligned} Y[0] &= 1 \\ Y[-3] &= Y[3] = -\frac{1}{2} \\ Y[k] &= 0 \ \text{ for } \ k \neq 0, \pm 3 \end{aligned}$$

Example-2 (3)

- $H(f)$ needs to eliminate the frequency components of $x(t)$ that do not exist in the output $y(t)$:

$$H(kf_0) = 0 \ \text{ for } \ k \neq 0, \pm 3$$

- To meet the above conditions, f_1 ,f_2 and f_3 must be chosen to pass the desired components and reject the undesired components. The desired components correspond to $0f_0 = 0$ and $3f_0 = \pm\frac{3}{4}$ Hz. Therefore, the following inequalities must be satisfied:

$$0 < f_1 < f_0 \quad \Rightarrow \quad 0 < f_1 < \frac{1}{4}$$

$$2f_0 < f_2 < 3f_0 \quad \Rightarrow \quad \frac{1}{2} < f_2 < \frac{3}{4}$$

$$3f_0 < f_3 < 4f_0 \quad \Rightarrow \quad \frac{3}{4} < f_3 < 1$$

Example-2 (4)

- The frequency response of the filter can be determined from:

$$Y[k] = H(kf_0)X[k] \quad \Rightarrow \quad H(kf_0) = \frac{Y[k]}{X[k]} \text{ for } X[k] \neq 0$$

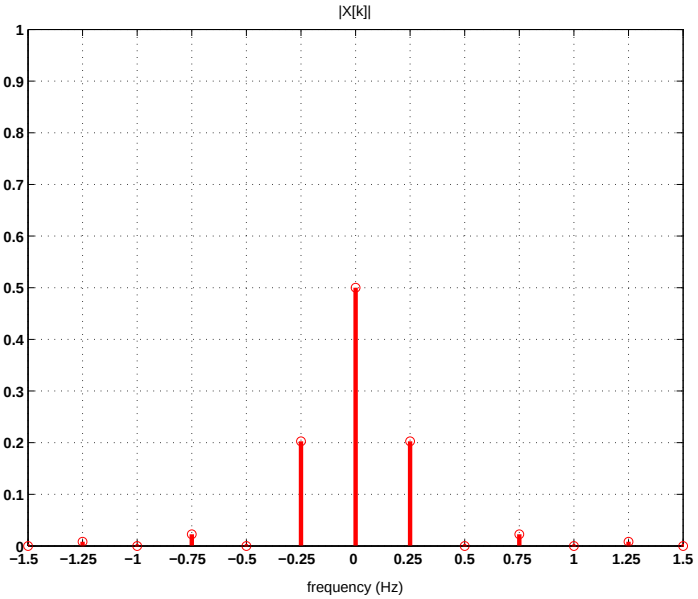
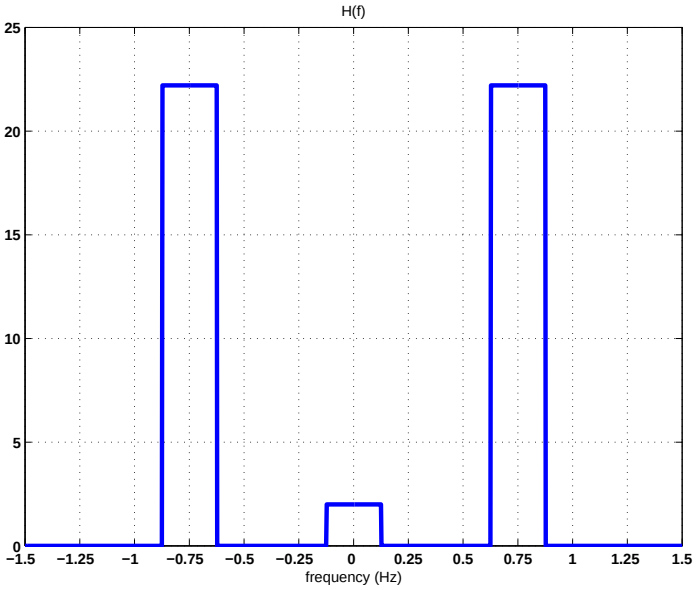
From the plot of the transfer function $H(f)$:

$$H(0f_0) = A_2 = \frac{Y[0]}{X[0]} = \frac{1}{\frac{1}{2}} = 2$$

$$H(-3f_0) = A_1 = \frac{Y[-3]}{X[-3]} = -\frac{1}{2X[-3]}$$

$$= -\frac{1}{2}\left[2\frac{\sin\left(3\frac{\pi}{2}\right)}{j(3\pi)^2}e^{-j3\frac{\pi}{2}}\right]^{-1} = -\frac{1}{2}\left[\frac{1}{2}\frac{j(3\pi)^2}{\sin\left(3\frac{\pi}{2}\right)}e^{j3\frac{\pi}{2}}\right] = 9\frac{\pi^2}{4}$$

$$H(3f_0) = A_3 = \frac{Y[3]}{X[3]} = -\frac{1}{2X[3]} = \cdots = 9\frac{\pi^2}{4}$$

Example-2 (5)

- The spectrum of the input signal and the frequency response of the filter (equalizer) are shown in the figure below: